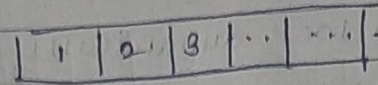
05/3/26

Arrays

Arrays is a list of elements of the same type placed in a contiguous memory location.



- contiguous

* Array declaration ↓

int [] arrayname ;

~~datatype [] arrname [] = new datatype [size]~~

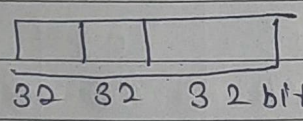
~~datatype arrname [] = { ... } ;~~

int [] rollnumber ;

* Array Definition

int rollnumber [] = new int [3] ;

datatype	name of array	keyword to allocate memory in heap	sized of array



- int.

eg : int x = 4

x → [4] 32 bit

eg : int [] arr = new int [3] ;

arr → [| |] 32 32 32

* Placing values in each block

rollno [101 | 102 | 103]
0 1 2

rollno [0] = 101 ;
rollno [1] = 102 ;
rollno [2] = 103 ;

* To retrive values in array

S.o.p.ln (rollno [0]) //
S.o.p.ln (rollno [1]) //

To avoid these manual work of placing values into array one by one & retrieving one by one we use → Loops

* using loops

```
int [] rollno = new int [3];
int x = 101;                    ~ To store 101
                                        102
                                        103
for (int i=0; i<3; i++)
{
    rollno [i] = x;             // store
    x++;
}

for (int i=0; i<3; i++)
{
    s.o.pln (rollno [i])        //print array
}
```

* If size of array is not known or to avoid hardcode

we use → nums.length
array name

```
for (i=0; i<nums.length; i++)
```

* If we try to retrieve any value from index, but that particular index is not here then, index out of Bound Exception occur

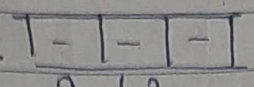
size = 3

1	2	3

Multi - Dimensional Array

1D - Array

int nums [] = new int [3];

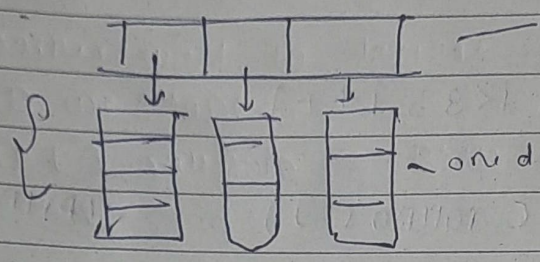


⇒ 1D Array - Array of int

contiguous memory & type same.

2D - Array

Array of Arrays



every cell has its own array & each array has int value (same type value)

eg: To store student Marks of 3 subj

	Hindi	Eng	CSE
S1	50	30	90
S2	60	40	80
S3	70	50	70

rows — no. of stud.

columns

int [] [] marks = new int [3] [3];

marks

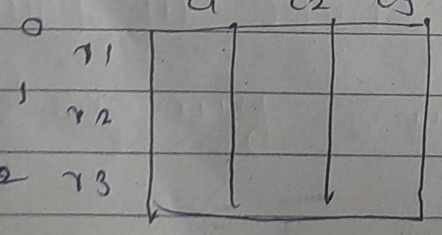
	H	E	M
S1	50	30	90
S2	60	40	80
S3	70	50	70

student

⇒ Internally it will look like matrix form

3 x 3

row col

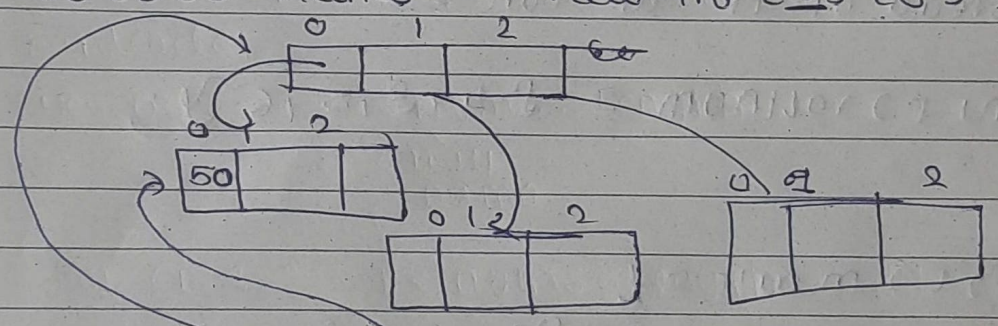


To print matrix → we use nested for loops

```
int marks [] [] = new int [3] [3];
marks [0] [0] = 50;
marks [ ] [ ] = 90
                ⋮
                                  or marks.length.
for (int rows = 0 ; rows < 3 ; row++)
{
    for (int cols = 0 ; cols < 3 ;  cols++)
    {                              marks [row].length
        S.O.P (marks [rows] [cols]);
    }
    S.O.P ln ();
}  }
```

* Conceptual Representation of 2D-array

int [] [] marks = new int [3] [3];



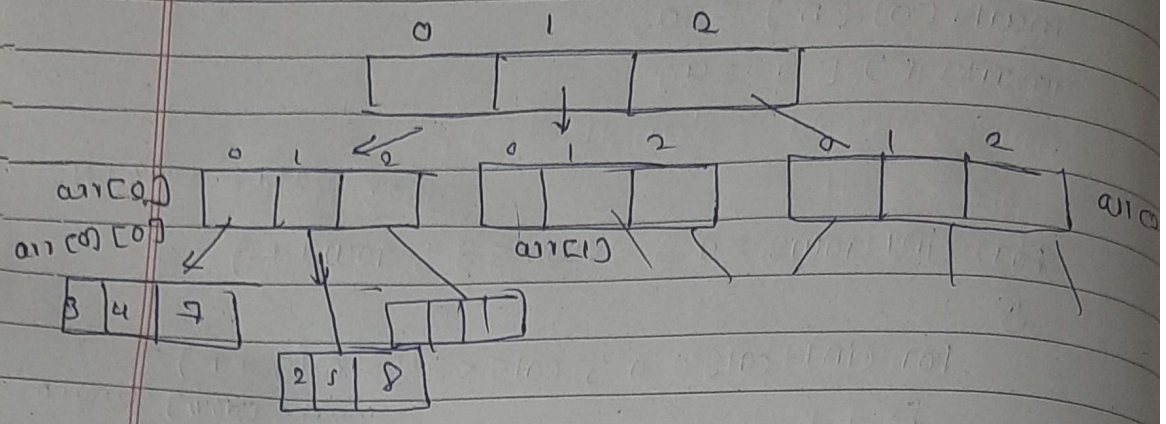
when marks [0] [0] = 50

represent main array

this represent array of array

3-D Array

int [][][] arr = new int [3][3][8];



Another way to declare array

1-D we done

int [] rollnum = new int [3];

rollno [0] = 4;

instead of it we can do

int [] rollnum = {4, 3, 5, 6};

2-D

int [] [] mark = {

{4, 5, 6}

{7, 8, 9}

};

Strings in Java

↳ Basically sequence of characters.

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* Declare string

String name = "sarali" ;

name → "sarali" - contains cells

S.O. Println (" ") ;
string text

To concatinate string we use + operator